React Questions & Answers

50 multiple-choice questions for practice

Question types: Single answer and multi-select

Choose the best option for single-answer questions. For multi-select questions, choose every correct option. The answer and a short explanation appear under each question.

Topics include JSX, components, props, state, hooks, events, forms, lists, Context, rendering, and accessibility.

Questions

01. What is React primarily used to build?

SINGLE ANSWER

A. Databases
B. User interfaces
C. Operating systems
D. Web servers

Correct answer: B
Why: React is a JavaScript library for building user interfaces.

02. Which syntax lets you write markup-like elements in JavaScript?

SINGLE ANSWER

A. SQL
B. JSX
C. YAML
D. GraphQL

Correct answer: B
Why: JSX is commonly used to describe React elements.

03. Which statement about React components is correct?

SINGLE ANSWER

A. They must be class-based
B. They can be reusable UI pieces
C. They can never receive data
D. They only run on servers

Correct answer: B
Why: Components let an interface be split into reusable pieces.

04. How does a parent normally pass data to a child component?

SINGLE ANSWER

A. Props
B. State setter only
C. CSS variables
D. URL fragments

Correct answer: A
Why: Props are inputs passed from a parent to a child.

05. Which hook adds local state to a function component?

SINGLE ANSWER

A. useRoute
B. useState
C. useStyle
D. useClass

Correct answer: B
Why: useState returns a value and a setter function.

06. What should you do to update state created by useState?

SINGLE ANSWER

A. Change the state variable directly
B. Call its setter
C. Edit the DOM manually
D. Reload the page

Correct answer: B
Why: Calling the setter asks React to render with the new state.

07. Which hook is intended for synchronizing with external systems?

SINGLE ANSWER

A. useEffect
B. useLabel
C. useMarkup
D. useRenderNow

Correct answer: A
Why: useEffect can run side effects such as subscriptions or network requests.

08. What is the usual purpose of a key in a rendered list?

SINGLE ANSWER

A. Set text color
B. Help React identify items between renders
C. Encrypt item data
D. Sort the list automatically

Correct answer: B
Why: Keys help React match list items when the list changes.

09. Which key is generally best for a list item?

SINGLE ANSWER

A. A stable unique ID
B. A random number generated every render
C. The item position in every case
D. The current time

Correct answer: A
Why: A stable ID preserves item identity across renders.

10. How do you attach a click handler in JSX?

SINGLE ANSWER

A. onclick="run()"
B. onClick={run}
C. click={run()}
D. on-click:run

Correct answer: B
Why: React event names use camelCase and receive a handler function.

11. What is a controlled input?

SINGLE ANSWER

A. An input whose value is driven by React state
B. An input hidden from React
C. An input with no value
D. An input controlled only by CSS

Correct answer: A
Why: Its displayed value comes from state and changes through an event handler.

12. What does lifting state up mean?

SINGLE ANSWER

A. Moving shared state to a common parent
B. Moving state into CSS
C. Copying state into every child
D. Deleting state

Correct answer: A
Why: A common parent can coordinate data shared by multiple children.

13. What is the purpose of a React Fragment?

SINGLE ANSWER

A. Group elements without an extra DOM wrapper
B. Fetch data
C. Create a route
D. Store global state

Correct answer: A
Why: Fragments let a component return multiple siblings without an extra element.

14. What does conditional rendering mean?

SINGLE ANSWER

A. Showing UI based on a condition
B. Rendering every component twice
C. Rendering only CSS
D. Changing the browser URL

Correct answer: A
Why: JavaScript conditions can choose which UI appears.

15. What does useContext help a component do?

SINGLE ANSWER

A. Read a value from a context provider
B. Create a database
C. Replace every prop
D. Compile JSX

Correct answer: A
Why: useContext reads the nearest matching provider value.

16. What is the purpose of useRef?

SINGLE ANSWER

A. Keep a mutable reference without causing a render when it changes
B. Replace all state
C. Force a component to render
D. Create a new component

Correct answer: A

Why: A ref keeps a value between renders; changing it does not trigger a render.

17. What does useMemo return?

SINGLE ANSWER

A. A memoized calculation result
B. A DOM node only
C. A network response automatically
D. A state setter

Correct answer: A

Why: useMemo can reuse a calculated value while dependencies are unchanged.

18. What does useCallback memoize?

SINGLE ANSWER

A. A function definition
B. A CSS selector
C. An HTML page
D. A component tree snapshot

Correct answer: A

Why: useCallback reuses a function reference while dependencies remain unchanged.

19. When can an effect cleanup function be useful?

SINGLE ANSWER

A. To unsubscribe or clear a resource
B. To change JSX into HTML
C. To reset the browser
D. To create props

Correct answer: A

Why: Cleanup can remove subscriptions, timers, or other resources.

20. What happens when a component state setter is called with a new value?

SINGLE ANSWER

A. React schedules an update and may render again
B. React edits source code
C. The browser closes
D. The component is permanently removed

Correct answer: A

Why: React schedules an update so the UI can reflect the new state.

21. Which statements about props are true?

MULTI-SELECT

A. They are passed into a component
B. They are read-only from the receiving component’s perspective
C. They can contain functions
D. A child should mutate them directly

Correct answer: A, B, C
Why: Props carry values or callbacks; a child should treat them as inputs.

22. Which are valid ways to create React components?

MULTI-SELECT

A. Function components
B. Class components
C. CSS files by themselves
D. Database tables

Correct answer: A, B
Why: React supports function components and class components.

23. Which statements about JSX are true?

MULTI-SELECT

A. It is transformed into JavaScript by tooling
B. Expressions can be embedded with braces
C. It requires every element to have a closing tag or be self-closing
D. It is a separate programming language executed directly by the browser

Correct answer: A, B, C
Why: JSX is syntax that build tooling transforms into JavaScript.

24. Which are valid list key choices when they are stable and unique?

MULTI-SELECT

A. A database ID
B. A stable slug
C. A random value regenerated on every render
D. An index that never changes because the list is static

Correct answer: A, B, D
Why: Stable unique identifiers are best; an index can work for an unchanging static list.

25. Which are common useEffect dependency behaviors?

MULTI-SELECT

A. No dependency array: after each committed render
B. Empty array: after mount, with development-mode nuances
C. Dependency array: re-run when a dependency changes
D. Dependencies are ignored by React

Correct answer: A, B, C
Why: The dependency array determines when React re-runs an effect.

26. Which tasks are suitable examples of effects?

MULTI-SELECT

A. Subscribe to a browser event
B. Connect to an external widget
C. Derive a simple value that can be calculated during render
D. Start a timer and clear it in cleanup

Correct answer: A, B, D
Why: Effects synchronize with external systems; simple derived values usually belong in render.

27. Which statements about state are true?

MULTI-SELECT

A. State belongs to a component or state-owning abstraction
B. Updating state can trigger a render
C. State should be mutated directly
D. State updates may be batched

Correct answer: A, B, D
Why: State is updated through React APIs, and React can batch updates.

28. Which are ways to handle a form submit in React?

MULTI-SELECT

A. Use onSubmit on a form
B. Call preventDefault when handling a client-side submit
C. Read controlled values from state
D. Use an attribute named on-submit instead of onSubmit

Correct answer: A, B, C
Why: React uses camelCase event props; preventDefault can stop a browser navigation.

29. Which statements about controlled components are true?

MULTI-SELECT

A. React state is the source of truth for the input value
B. An onChange handler commonly updates state
C. They can make validation easier to coordinate
D. They cannot be used with forms

Correct answer: A, B, C
Why: Controlled fields make values explicit and easier to coordinate with validation.

30. Which statements about React event handlers are true?

MULTI-SELECT

A. They are commonly passed as functions
B. Event prop names use camelCase
C. Calling a handler during render is always the same as passing it
D. They can update state

Correct answer: A, B, D
Why: Pass a function reference such as onClick={handleClick}; use state setters for updates.

31. Which statements about Context are true?

MULTI-SELECT

A. A provider supplies a value to descendants
B. useContext can read the nearest provider value
C. Context is useful for some shared values
D. Every state value must be placed in Context

Correct answer: A, B, C
Why: Context helps share selected values without passing props through every layer.

32. Which statements about useRef are true?

MULTI-SELECT

A. A ref object persists across renders
B. Changing ref.current alone does not trigger a render
C. Refs can point to DOM elements
D. Every ref update automatically re-renders the component

Correct answer: A, B, C
Why: Refs are persistent mutable containers and can also connect to DOM nodes.

33. Which statements about useMemo and useCallback are true?

MULTI-SELECT

A. useMemo caches a value calculation
B. useCallback caches a function reference
C. They should be used only when there is a reason
D. They make every component faster automatically

Correct answer: A, B, C
Why: Memoization has tradeoffs and should address a real recalculation or identity concern.

34. Which statements about rendering are true?

MULTI-SELECT

A. A state update can cause rendering
B. A parent render can render its children again
C. Rendering should remain free of side effects
D. A render always means the DOM will change

Correct answer: A, B, C
Why: React can render to determine changes; rendering does not guarantee a DOM mutation.

35. Which statements about component props are true?

MULTI-SELECT

A. A component can receive strings
B. A component can receive arrays or objects
C. A component can receive callback functions
D. Props automatically become mutable local state

Correct answer: A, B, C
Why: Props can carry many JavaScript values, including callbacks.

36. Which statements about conditional UI are true?

MULTI-SELECT

A. An if statement can select a returned component
B. A ternary can select between two UI branches
C. Logical && can conditionally show an element
D. Conditional rendering requires a separate page

Correct answer: A, B, C
Why: JavaScript expressions and statements can choose which elements a component returns.

37. Which statements about React keys are true?

MULTI-SELECT

A. Keys should be unique among siblings
B. Keys help React track list items
C. A key is available as a normal prop automatically
D. Changing a key can cause React to treat an element as a different instance

Correct answer: A, B, D
Why: Keys guide reconciliation and are special metadata, not ordinary props.

38. Which statements about effect cleanup are true?

MULTI-SELECT

A. Cleanup can run before an effect re-runs
B. Cleanup can run when a component is removed
C. Cleanup can clear a timer
D. Cleanup is a place to return JSX

Correct answer: A, B, C
Why: Effects may return a cleanup function to release resources.

39. Which approaches can share data between sibling components?

MULTI-SELECT

A. Lift state to their common parent
B. Use a suitable context provider
C. Make each sibling keep unrelated duplicate values and hope they stay synced
D. Pass values and callbacks through props

Correct answer: A, B, D
Why: A parent, Context, or props can coordinate shared data.

40. Which statements about React fragments are true?

MULTI-SELECT

A. They group sibling elements in a return value
B. They add no wrapper DOM element
C. The short syntax <>...</> cannot receive a key
D. The explicit Fragment form can receive a key

Correct answer: A, B, C, D
Why: Use React.Fragment with a key when a keyed fragment is needed in a list.

41. Which statements about state setter functions are true?

MULTI-SELECT

A. They can accept a replacement value
B. They can accept an updater function based on previous state
C. They should be called during an event or other appropriate logic
D. They mutate the state variable immediately in place

Correct answer: A, B, C
Why: Updater functions are useful when the next value depends on the previous value.

42. Which statements about data fetching in React are generally sound?

MULTI-SELECT

A. Handle loading and error states
B. Avoid setting state after an obsolete request when appropriate
C. Consider framework or data-library features for larger apps
D. A fetch effect always runs exactly once in every environment

Correct answer: A, B, C
Why: Robust fetching handles lifecycle and status; development behavior can expose repeated effects.

43. Which are good accessibility practices for React forms?

MULTI-SELECT

A. Associate labels with inputs
B. Use semantic button elements
C. Provide clear validation feedback
D. Use placeholder text as the only label

Correct answer: A, B, C
Why: Labels, semantic controls, and feedback help users and assistive technology.

44. Which statements about component composition are true?

MULTI-SELECT

A. Components can be nested
B. A component can receive children
C. Composition can avoid overly complex components
D. Every component must render a single HTML tag

Correct answer: A, B, C
Why: Composition builds larger interfaces from smaller components and children.

45. Which are reasons to split a UI into components?

MULTI-SELECT

A. Reuse a repeated interface
B. Give a logical part a clear responsibility
C. Make every one-line expression a separate component
D. Keep related UI and behavior together

Correct answer: A, B, D
Why: Useful component boundaries improve reuse and clarity without unnecessary fragmentation.

46. Which statements about StrictMode in development are true?

MULTI-SELECT

A. It can help reveal unsafe patterns
B. It may intentionally re-run certain logic in development
C. It is a production performance optimization switch
D. It can help surface missing effect cleanup

Correct answer: A, B, D
Why: StrictMode adds development checks; it does not change production behavior in the same way.

47. Which statements about state and derived values are true?

MULTI-SELECT

A. A value calculable from current props and state can often be calculated during render
B. Duplicating derived data in state can create synchronization bugs
C. useMemo may help cache an expensive calculation
D. Every derived value must use useEffect

Correct answer: A, B, C
Why: Calculate simple derived values directly; memoize only when useful.

48. Which statements about forms and event handling are true?

MULTI-SELECT

A. onChange can update an input state value
B. onSubmit can handle form submission
C. A submit button inside a form can submit it by default
D. React events require manually adding inline HTML strings

Correct answer: A, B, C
Why: React event props use JavaScript handlers rather than HTML string handlers.

49. Which statements about hooks are true?

MULTI-SELECT

A. Hooks are called at the top level of function components or custom hooks
B. Hooks should not be called conditionally
C. Custom hooks can reuse stateful logic
D. A hook name often starts with use

Correct answer: A, B, C, D
Why: Following hook rules keeps hook call order consistent across renders.

50. Which practices help keep React components understandable?

MULTI-SELECT

A. Use descriptive names
B. Keep responsibilities focused
C. Extract repeated behavior when it improves clarity
D. Put all application logic in one giant component

Correct answer: A, B, C
Why: Clear names, focused components, and purposeful reuse improve maintainability.